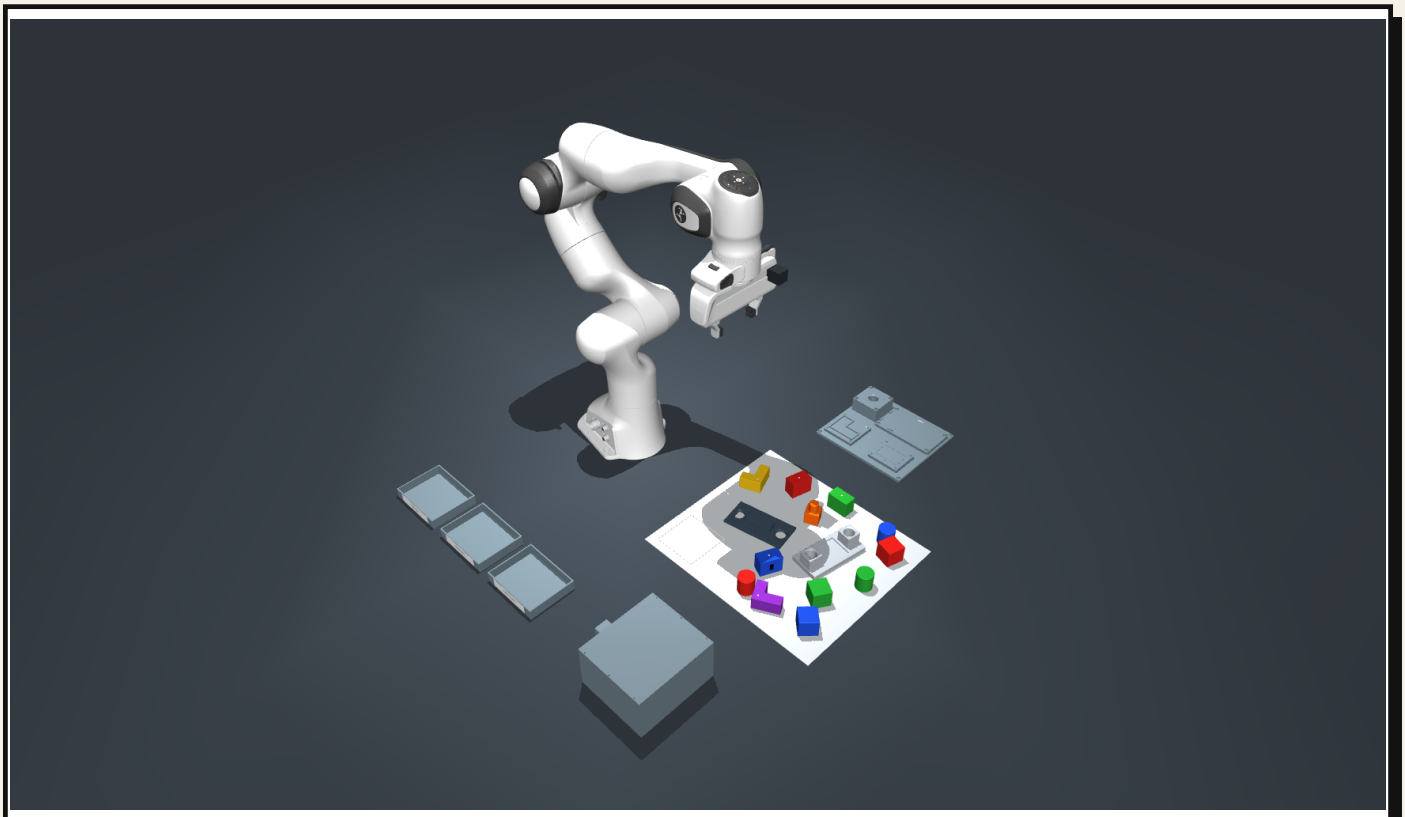
FRANKA EMIKA PANDA / ROS 2 / CHALLENGE DESCRIPTION

PANDA MANIPULATION CHALLENGE



MuJoCo simulation / Panda model: MuJoCo Menagerie (Apache-2.0)

Develop a ROS 2 application that enables the Franka Emika Panda to complete six manipulation tasks using perception, motion planning, grasping and task sequencing. Develop and test your methods in simulation for transfer to the physical workspace.

Complete Levels 1-6 in order without resetting the workspace between levels. Starting poses may vary between attempts. A wrist-mounted camera and an external camera overlooking the table are available; use either camera or both. The final task is to assemble a five-part product.

01 PICK | 02 SORT | 03 STACK | 04 REGRASP | 05 INSERT | 06 ASSEMBLE

WORKSPACE

Setup

REFERENCE TOP VIEW

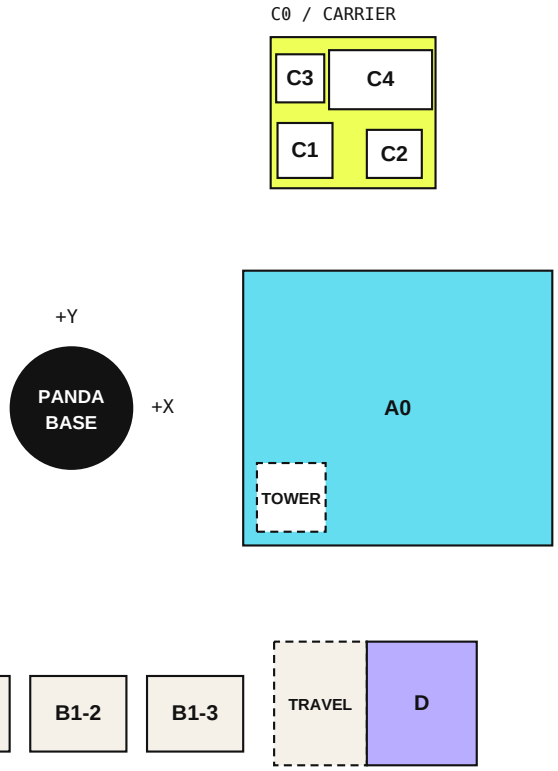


TABLE PLANE / ROBOT BASE ORIGIN

A0

Staging, tower and temporary table support

B1-1/2/3

Cube / cuboid / cylinder trays

C0

Fixture carrier

C1

L-piece skill fixture

C2

Marked cuboid target

C3

Round test socket

C4

Product assembly nest

D

Drawer and handle

TRAY ASSIGNMENTS

CHECKPOINT	B1-1	B1-2	B1-3
01 / colour routing	Red cube	Green cube	Blue cube
02 / shape sorting	All cubes	All cuboids	All cylinders

INITIAL STATE AND CAMERAS

14 parts on A0 + O6-1 inside closed D1. The tower destination and all target fixtures start empty. No reset or relabelling between levels.

D0 case / D1 drawer / D2 handle / D3 guides. Use the wrist-mounted camera, external camera or both. Open the drawer in Level 6.

WORKPIECES

Workpiece overview

CUBES

01

3 parts. Red, green, blue.



01-R / 01-G / 01-B

40 × 40 × 40 mm

CUBOIDS

02

3 parts. White top; black front.



02-R / 02-G / 02-B

50 × 30 × 40 mm

CYLINDERS

03

3 parts. Flat ends; start upright.



03-R / 03-G / 03-B

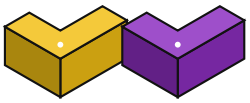
Diameter 36 / height 40 mm

L-SHAPED PIECES

04



2 parts. Yellow and violet; white top mark.



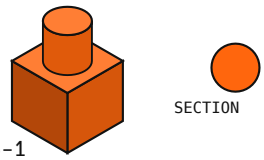
04-Y / 04-V

60 × 50 L-footprint / 30 mm thick

ROUND PEG

05

1 part. Orange; circular shaft.



05-1

Head 32 × 32 × 30 / shaft Ø20 × 20 mm

D-SHAPED PEG

06

1 part. Turquoise; one shaft flat. Starts in D1.



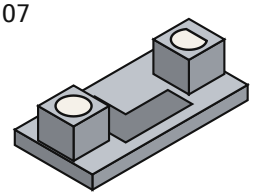
06-1

Same head / D-shaft 20 mm long

STEPPED BASE

07

1 part. L-recess and two integral peg bosses.



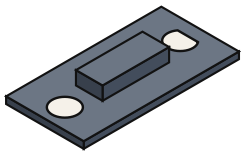
07-1

P1 base / light grey

COVER + GRIP BAR

08

1 part. Round + D holes; integral top grip bar.



08-1

P1 cover / dark grey

SCHEMATIC VIEWS / L-PLAN AND ROUND / D-SHAFT SECTIONS SHOWN

COMPETITION RULES

Information

01 NO RESETS BETWEEN LEVELS

Complete the levels in order. Each level starts from the state left by the previous level. Do not reset, respawn, reload or manually rearrange the workspace between levels.

02 AUTONOMY & PERCEPTION

Run autonomously through ROS 2. After the start signal, report every human intervention. No workpiece poses are supplied. Use observations from either or both cameras and your action history; privileged simulator state, generated scene files, instructor files and random seeds are forbidden as object-pose inputs.

03 KEEP PARTS & CHECKPOINTS

Preserve parts and completed placements. All fifteen parts are present from the beginning. Leave unrelated parts in stable positions. Keep completed results intact unless a later task explicitly reuses them: Level 2 re-sorts the cubes; Level 6 removes O4-Y from C1 and O5-1 from C3. The Level 3 tower and O2-B in C2 must remain intact through the finish.

04 USE FREE TABLE SUPPORT

Keep equipment fixed and travel clear. Free A0 surface may support temporary placement and regrasping. Keep the tower destination clear until Level 3 and leave drawer travel unobstructed throughout. Avoid unintended collisions, forceful wedging and losing parts; intended grasping, support, insertion and drawer contacts are allowed.

05 PLACEMENT ACCEPTANCE

Release and verify stability. A checkpoint counts only after all relevant parts have been released and remain stable without gripper support for at least two seconds. Tray and marked-area placements must fit inside their boundaries; parts may extend above tray rims. Tower layers must be centred over their supports, with no contact to other equipment.

06 REQUIRED OPERATIONS

Demonstrate the required operations. Record the Level 4 release and change of opposing gripping faces. Keep D1 closed and O6-1 inside it through Level 5. In Level 6, open D1 using D2, retrieve O6-1 and close D1 again. A checkpoint remains credited when its result is legitimately reused later.

07 GRASPING AND ASSEMBLY

Use physical grasps and seated joints. Parts must be held by the simulated or physical gripper. Artificial attachment, teleportation and physics/state edits are forbidden; MoveIt planning attachments may represent a real grasp. P1 must stand assembled in C4 without gripper support; no lift test is required for its gravity-seated joints.

LEVEL 01 OF 06

1 2 3 4 5 6

01 LOCATE, PICK & PLACE

LOCATE WORKPIECES FROM CAMERA OBSERVATIONS

STARTING STATE

The complete fifteen-part inventory is present. Fourteen parts are on A0; O6-1 is inside the closed drawer. Locate O1-R, O1-G and O1-B from observations.

YOUR TASK

Grasp and transport the three cubes according to this colour-routing instruction. Keep the other parts in place and leave the tower destination clear.

Workpiece	Destination
O1-R / red cube	B1-1
O1-G / green cube	B1-2
O1-B / blue cube	B1-3

Tray IDs and their permanent shape labels stay unchanged. The mixed tray contents are intentional preparation for Level 2.

COMPLETED WHEN

All three cubes are stable inside their assigned trays. The other twelve parts remain present; the drawer is closed.

FOCUS / Cube localization, coordinate transforms, grasp planning and gripper control.

WORKFLOW

APPROACH → **GRASP** → **TRANSPORT** → **RELEASE**

LEVEL 02 OF 06

1 2 3 4 5 6

02 PERCEIVE & SORT BY SHAPE

LOCATE WORKPIECES FROM CAMERA OBSERVATIONS

STARTING STATE

Continue from Level 1. The cubes are distributed across the trays; all six cuboids and cylinders remain on A0. The other parts retain their current positions.

YOUR TASK

Use one or both cameras to identify the nine O1/O2/O3 parts and determine their current poses. Sort them by shape, regardless of colour:

Workpieces	Destination
O1-R / O1-G / O1-B - cubes	B1-1
O2-R / O2-G / O2-B - cuboids	B1-2
O3-R / O3-G / O3-B - cylinders	B1-3

Move the green and blue cubes out of their Level 1 trays. A correctly placed part may stay where it is. Leave O4-O8 available and keep the tower destination clear.

COMPLETED WHEN

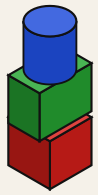
Each tray contains its three correct parts. No O1/O2/O3 part remains on A0 or in the wrong tray. D1 remains closed.

FOCUS / Pose estimation, shape classification and action ordering. RGB and depth are available; using depth is optional.

WORKFLOW

OBSERVE → **CLASSIFY** → **GRASP** → **SORT**

03 SELECT & STACK THE ORDER TOWER



120 mm

LOCATE WORKPIECES FROM CAMERA OBSERVATIONS

STARTING STATE

Continue from the shape-sorted trays. The tower destination on A0 is clear; the remaining workpieces stay in the workspace.

YOUR TASK

Retrieve the requested parts from their trays and build this three-layer tower at the marked destination on A0:

Layer / build order	Required workpiece
1 / bottom, resting on the table	O1-R / red cube
2 / middle, white top mark upward	O2-G / green cuboid
3 / top, cylinder axis vertical	O3-B / blue cylinder

Centre each layer over the one below it. The cuboid and cylinder may overhang their supports because their nominal widths differ; their centre projections must align within 5 mm in each horizontal axis. Release after each placement.

COMPLETED WHEN

The complete 120 mm nominal-height tower stands unsupported for two seconds, inside its marked destination. All non-target parts remain stable; keep this tower intact through Levels 4-6.

FOCUS / Selective retrieval, stacking, centre alignment and stability verification.

WORKFLOW

SELECT → **CUBE** → **CUBOID** → **CYLINDER**

04 ORIENT, REGRASP & PLACE

LOCATE WORKPIECES FROM CAMERA OBSERVATIONS

STARTING STATE

Continue with the tower intact. O4-Y and O4-V are still on A0. O2-B is in B1-2; C1 and C2 are empty.

YOUR TASK

Complete both operations:

1. Pick the yellow L-piece **O4-Y** and fit it into **C1**, with the white top mark upward. Leave the violet L-piece O4-V as the distractor on A0.

2. Retrieve the blue cuboid **O2-B**. Release it on free, level surface within **A0**. Regrasp it using a **different pair of opposing side faces**, switching between its nominal 30 mm and 50 mm gripping directions. Place it in **C2**, white top upward and black front mark facing the target arrow.

A wrist rotation without releasing and changing gripping faces does not count.

COMPLETED WHEN

O4-Y is seated in C1, O2-B is correctly oriented in C2 and the release/regrasp has been demonstrated. The tower remains intact and D1 closed.

FOCUS / Orientation constraints, grasp changes and preservation of completed placements.

WORKFLOW / CUBOID REGRASP

GRASP → **RELEASE ON A0** → **REGRASP** → **PLACE IN C2**

LEVEL 05 OF 06

05 REORIENT THE ROUND PEG

LOCATE WORKPIECES FROM CAMERA OBSERVATIONS

STARTING STATE

Continue from Level 4. O5-1 is still on A0, resting on its head with the shaft upward. C3 is empty. O6-1 stays inside the closed drawer.

YOUR TASK

Grasp the orange round peg **O5-1** by its head, reorient its shaft downward and insert it into the round socket **C3**. Use free A0 surface for an intermediate release or regrasp if needed.

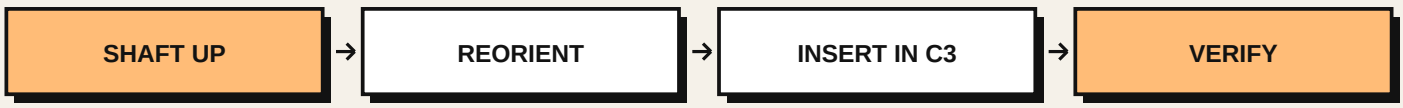
Approach in alignment with the socket, control the insertion and verify that the head underside reaches the socket's top surface. Leave the peg accessible for withdrawal in Level 6. Do not open the drawer or use O6-1 at this level.

COMPLETED WHEN

O5-1 is fully seated in C3 and stable after release. A peg balanced on the rim does not count. The tower, C1 placement and C2 placement remain intact.

FOCUS / Out-of-plane reorientation, controlled insertion and seating verification.

WORKFLOW



06 ASSEMBLE PRODUCT P1

P1 ASSEMBLY DRAWING / PAGE 11

STARTING STATE

Continue from Level 5. O4-Y is in C1, O5-1 is in C3, and O7-1 and O8-1 remain on A0. O6-1 is still in closed D1. The product nest C4 is empty.

YOUR TASK

Build **P1, the pinned locator cassette**, as one five-part assembly. Use the target product illustration supplied with the challenge. Retrieve and reuse parts from earlier checkpoints:

Assembly operation	Source and required result
Locate O7-1 / stepped base	A0 to C4; match the nest's orientation mark.
Fit O4-Y / yellow L-piece	Withdraw from C1; seat in the base's L-recess, white top upward.
Place O8-1 / cover	A0 onto the base bosses; grip bar upward and both holes aligned.
Insert O5-1 / round peg	Withdraw from C3; insert through the cover's round hole into the base.
Retrieve and insert O6-1 / D-peg	Open D1 using D2; retrieve the peg, turn its shaft downward and align both D-profiles before insertion.

Seat both peg heads against the cover and close D1 after retrieval. C1 and C3 may now be empty. Keep the Level 3 tower, O2-B in C2, remaining sorted parts and violet distractor intact.

COMPLETED WHEN

All five P1 parts are correctly fitted together in C4 and stable after release; both peg heads are seated, D1 is closed and the protected earlier results are intact.

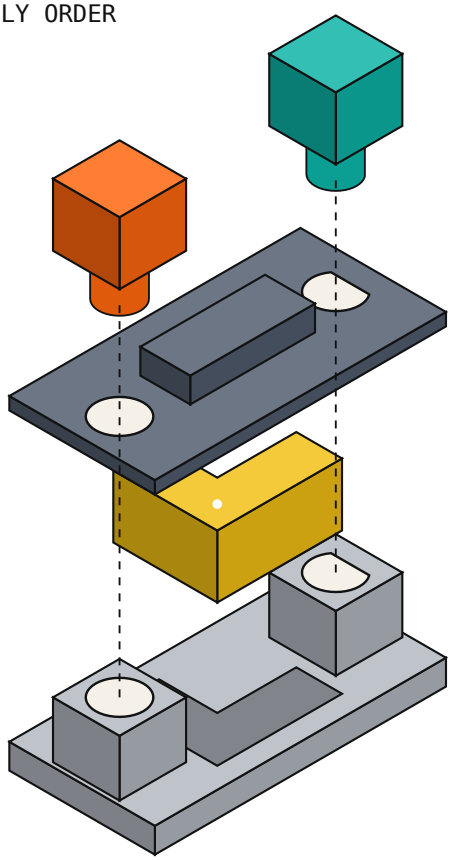
FOCUS / Multi-part assembly, dependency planning, extraction of previously inserted parts, keyed insertion, drawer operation and recovery.

LEVEL 06 / PRODUCT DRAWING

Product P1

Five-part locator cassette. C4 supports the base during assembly of the insert, cover and both pegs. The top grip bar is part of O8-1.

EXPLODED VIEW / ASSEMBLY ORDER



04
O5-1 / ROUND PEG

05
O6-1 / D-PEG

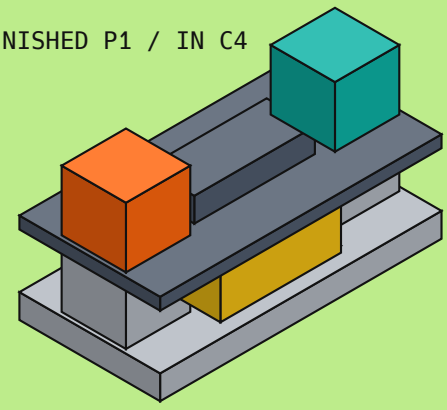
03
O8-1 / COVER

02
O4-Y / L-INSERT

01
O7-1 / BASE

Both pins pass through the cover into the base. Match the D-shaped flats in all three parts.

FINISHED P1 / IN C4



5 COMPONENTS / NO GRIPPER SUPPORT

Acceptance checklist

- ☐ O7 sits correctly in C4; O4-Y is inside its recess, white mark up.
- ☐ O8 rests on the base bosses; its grip bar faces upward.
- ☐ Both peg heads seat against O8. D1 is closed.
- ☐ Tower and C2 result remain intact. No product lift test is required.